

6.4 CALCULATING LONGITUDINAL CG (see Fig. 6-2)

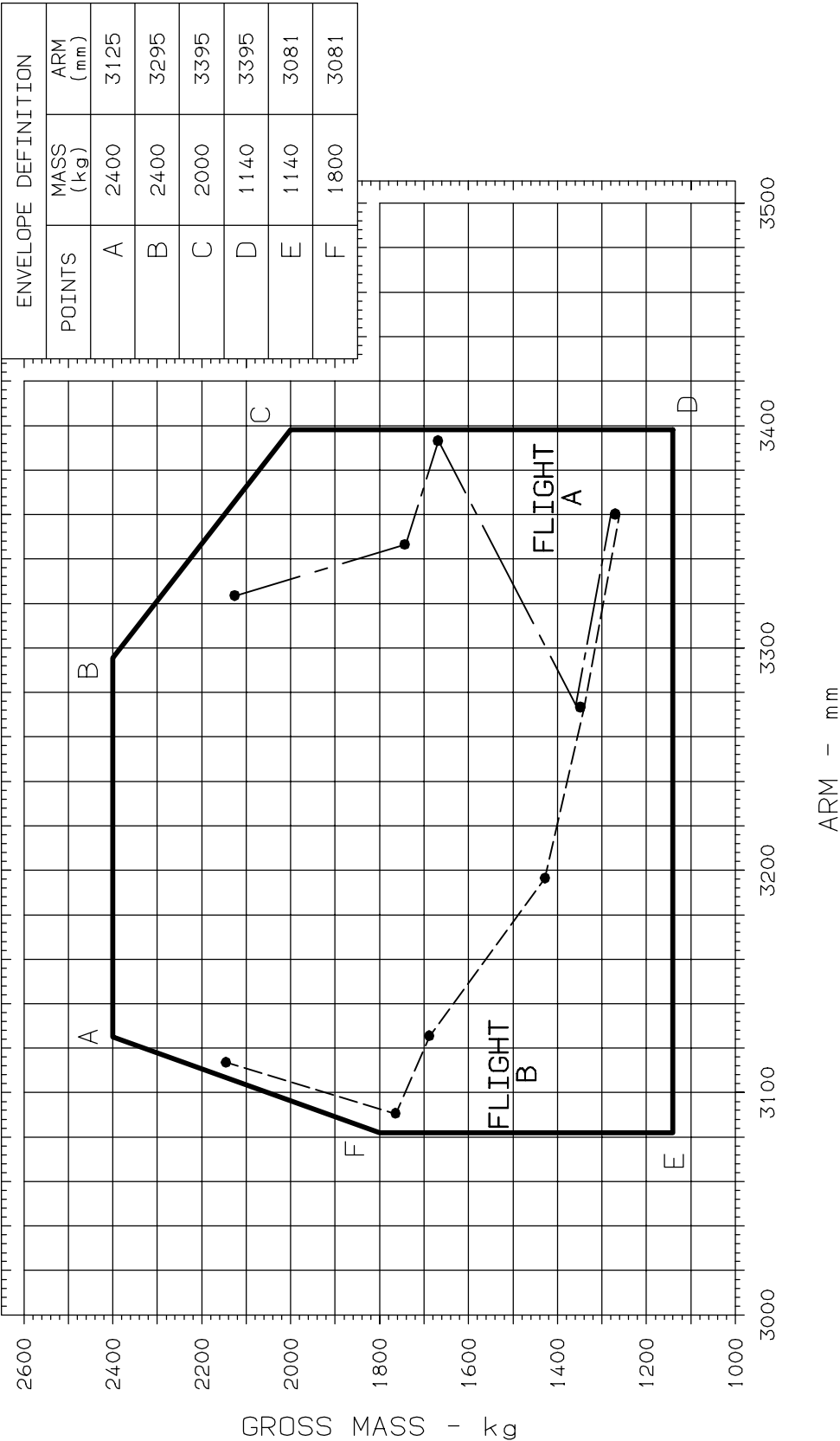
The following examples show a method of calculating the longitudinal center of gravity.

NOTE Basic empty mass data are to be taken from the latest entry on the MASS AND BALANCE RECORD, Form MBR-1 appended to this Section.

LOADING EXAMPLES						
Flight A: 1 pilot (80 kg), bulk cargo load (320 kg), and full fuel (456 kg). The resultant CG is solved as follows:						
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Resultant		
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Basic empty mass	1300	3375	4387500	1300	3375	4387500
Pilot	80	1650	132000	1380	3275	4519500
Bulk cargo	320	3900	1248000	1700	3393	5767500 ¹⁾
Fuel, supply tank	74	2275	168350	1744	3346	5935850 ²⁾
Fuel, main tank	382	3220	1230040	2156	3324	7165890 ³⁾
Flight B: 2 pilots (2 x 80 kg), 3 passengers (3 x 80 kg), and full fuel (456 kg). The resultant CG is solved as follows:						
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Resultant		
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Basic empty mass	1300	3375	4387500	1300	3375	4387500
Pilot and copilot	160	1650	264000	1460	3186	4651500
Passengers (3)	240	2740	657600	1700	3123	5309100 ¹⁾
Fuel, supply tank	74	2275	168350	1774	3088	5477450 ²⁾
Fuel, main tank	382	3220	1230040	2156	3111	6707490 ³⁾
NOTE To make certain that the CG does not "wander" outside of the CG envelope during the flight, the helicopter must be so loaded that the above points ¹⁾ (minimum landing mass), ²⁾ (supply tank full), and ³⁾ (supply and main tanks full) lie within the approved limits.						

NOTE The lateral CG position will always remain within limits regardless of the location of internal loads provided that the gross mass remains within limitations.

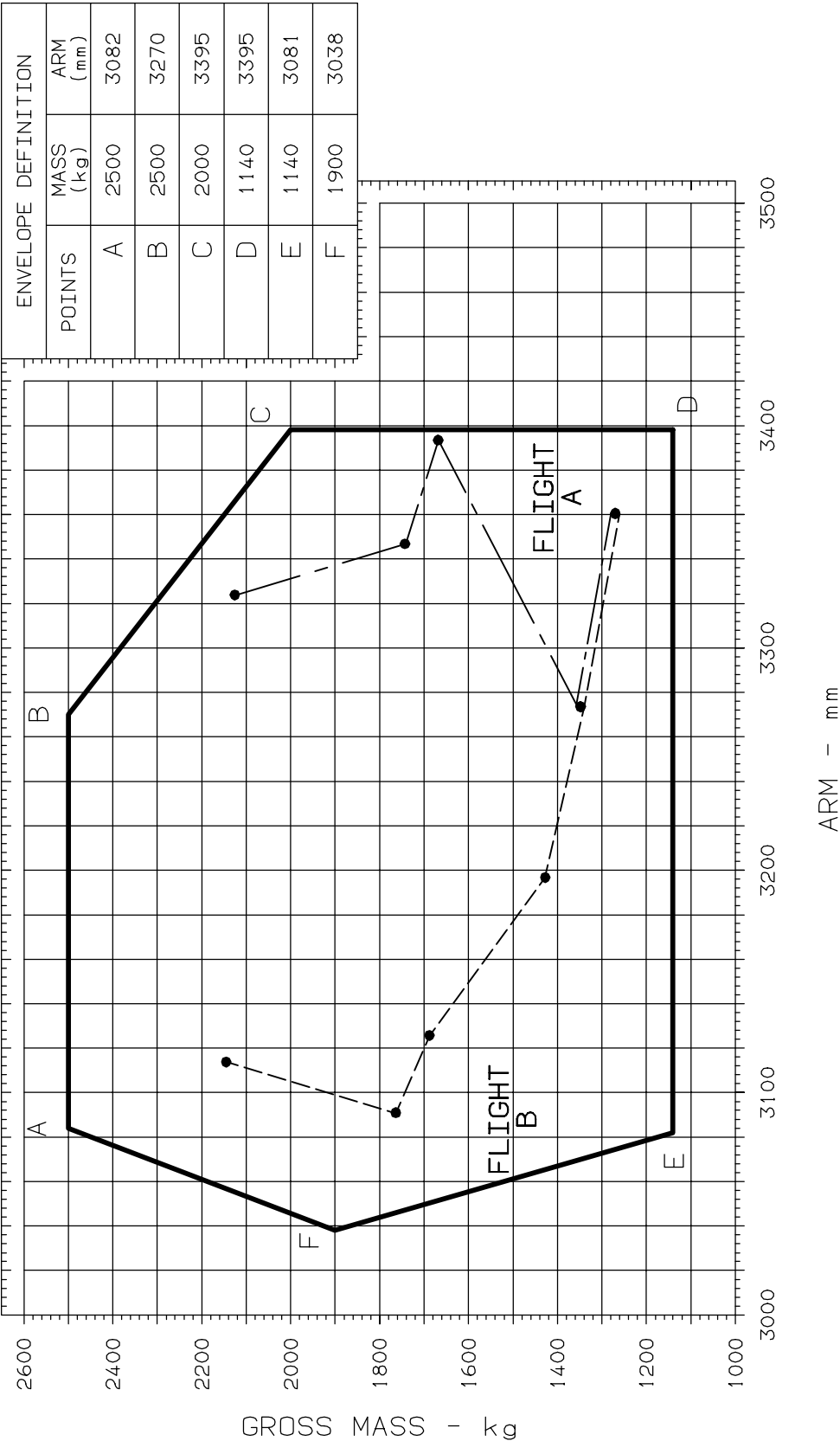
BO5-6BEM,1



NOTE REFER TO TEXT FOR LOADING EXAMPLE DATA.

Fig. 6-2 CG Envelope with Loading Examples

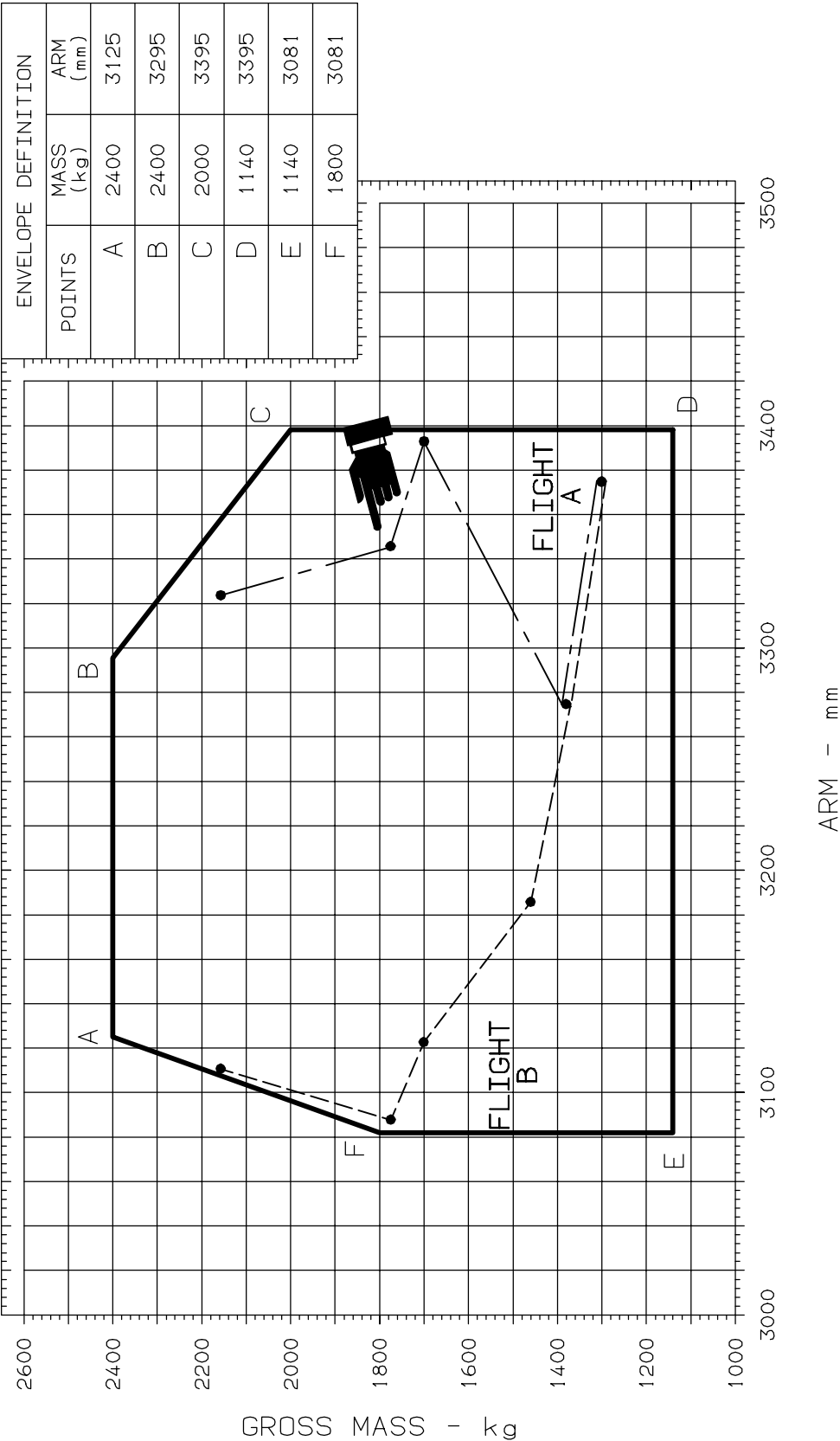
BO5-6BGM,1



NOTE REFER TO TEXT FOR LOADING EXAMPLE DATA.

Fig. 6-2 CG Envelope with Loading Examples

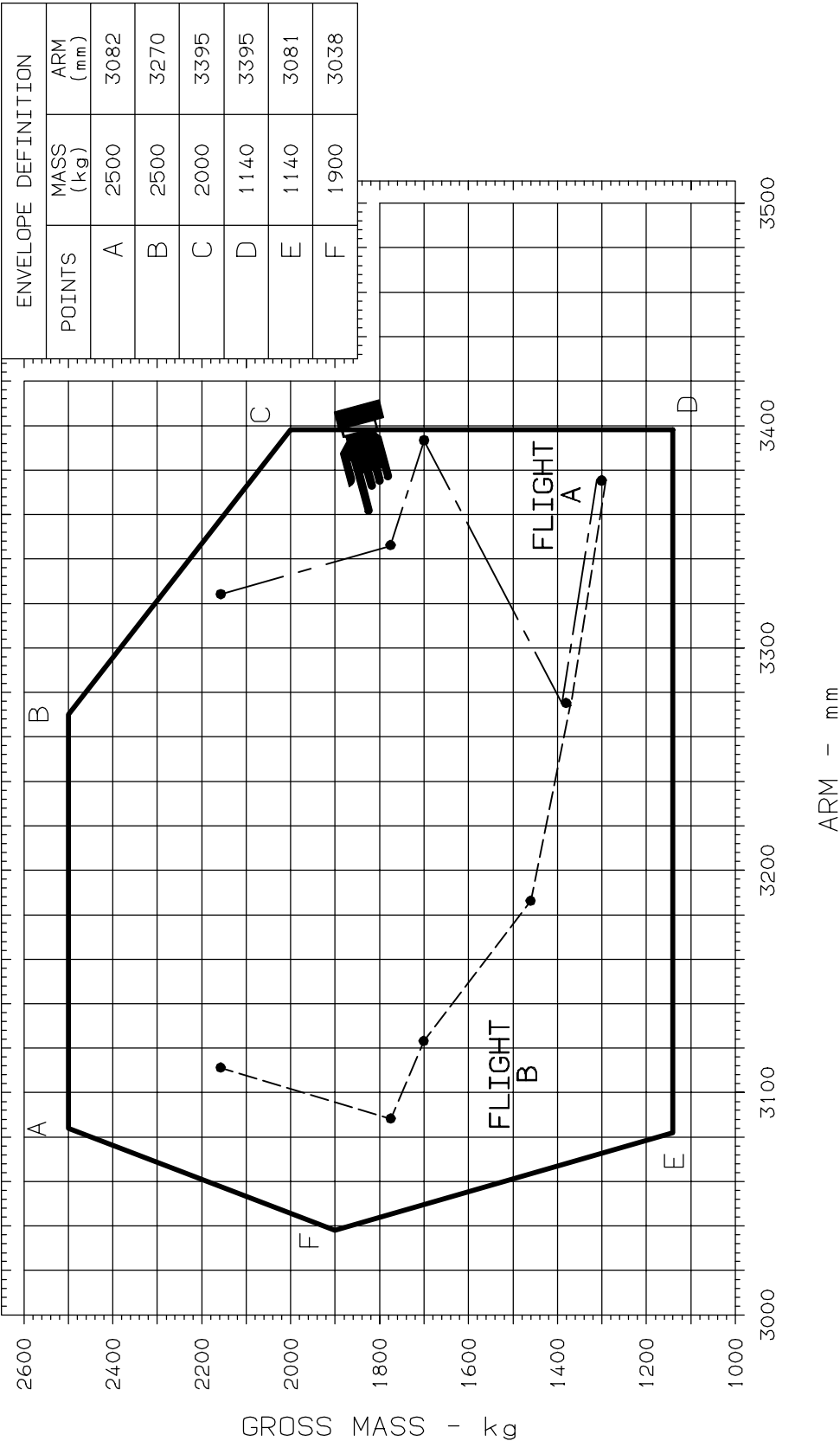
BO5-6BJM,1



NOTE REFER TO TEXT FOR LOADING EXAMPLE DATA.

Fig. 6-2 CG Envelope with Loading Examples

BO5-6BLM,1



NOTE REFER TO TEXT FOR LOADING EXAMPLE DATA.

Fig. 6-2 CG Envelope with Loading Examples

